

## SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

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### COURSE OUTCOME COVERED

**CO3:** Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)

Assessment Tool Weightage: 10%

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QUESTIONS: 5

TOTAL MARKS: 50

Annexure A  
MCQ

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### Code of Conduct:

I T. Lakshmi prasad Naidu (192324187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Lakshmi prasad Naidu

**Annexure A**  
**MCQ**

**MCQ (Additional Set)**

1. Which property ensures the same input always produces the same hash?  
A) Avalanche effect  
☒ B) Determinism  
C) Collision resistance  
D) Diffusion
2. What is the primary purpose of SHA-3?  
A) Faster encryption  
☒ B) Alternative secure hashing standard  
C) Data compression  
D) Key exchange
3. Which attack tries all possible inputs to match a hash?  
A) Replay attack  
☒ B) Brute-force attack  
C) MITM attack  
D) DoS attack
4. What is a salted hash?  
A) Encrypted hash  
☒ B) Hash with random data added  
C) Shortened hash  
D) Reversed hash
5. Which algorithm is widely used for password hashing?  
A) AES  
B) RSA  
☒ C) bcrypt  
D) DES
6. What does second pre-image resistance ensure?  
A) Same input produces same output  
☒ B) Hard to find different input with same hash  
C) Fast hashing  
D) Small hash size
7. What is the output size of SHA-512?  
A) 256 bits  
B) 128 bits  
☒ C) 512 bits  
D) 1024 bits
8. Which technique slows down brute-force attacks?  
A) Encryption  
B) Key exchange  
☒ C) Key stretching  
D) Compression
9. What is the purpose of a nonce?  
A) Encryption  
☒ B) Prevent replay attacks  
C) Compress data  
D) Generate keys
10. Which function is irreversible?  
A) Encryption  
☒ B) Hashing

- C) Compression
- D) Encoding
- 11. Which algorithm is used in DSS?
  - A) AES
  - ☒ B) DSA
  - C) DES
  - D) RC4
- 12. What is required to verify a digital signature?
  - A) Private key
  - ☒ B) Public key
  - C) Session key
  - D) Shared key
- 13. What does a digital signature guarantee?
  - A) Confidentiality
  - ☒ B) Integrity and authenticity
  - C) Compression
  - D) Speed
- 14. What happens if message digest changes?
  - A) Signature remains valid
  - ☒ B) Verification fails
  - C) Encryption fails
  - D) Speed increases
- 15. Which attack involves forging a signature?
  - A) Replay
  - ☒ B) Forgery
  - C) DoS
  - D) Sniffing
- 16. What is the role of a hash in signatures?
  - A) Encrypt data
  - B) Reduce size
  - ☒ C) Ensure integrity
  - D) Store keys
- 17. Which key is kept secret in signatures?
  - A) Public key
  - ☒ B) Private key
  - C) Session key
  - D) Shared key
- 18. What ensures trust in certificates?
  - A) Encryption
  - ☒ B) Certificate Authority
  - C) Compression
  - D) Hash size
- 19. What is certificate revocation?
  - A) Key generation
  - ☒ B) Canceling validity
  - C) Encryption
  - D) Compression
- 20. What is OCSP used for?
  - A) Encryption
  - ☒ B) Certificate validation
  - C) Compression
  - D) Routing
- 21. What is the main goal of Kerberos tickets?
  - A) Encryption
  - ☒ B) Authentication

- C) Signature
- D) Hash
- 33. What ensures certificate integrity?
  - A) Encryption
  - ☒ B) Digital signature
  - C) Compression
  - D) Routing
- 34. What is a root certificate?
  - A) Expired certificate
  - ☒ B) Self-signed trusted certificate
  - C) Invalid certificate
  - D) Temporary certificate
- 35. What is certificate pinning?
  - A) Storing keys
  - ☒ B) Binding certificate to host
  - C) Encryption
  - D) Compression
- 36. What is the risk of weak CA validation?
  - A) Faster speed
  - ☒ B) MITM attacks
  - C) Compression
  - D) Routing
- 37. What is the purpose of trust chain?
  - A) Encryption
  - ☒ B) Validate certificate hierarchy
  - C) Compress data
  - D) Routing
- 38. What is digital certificate used for?
  - A) Compression
  - ☒ B) Authentication
  - C) Routing
  - D) Storage
- 39. What ensures non-repudiation?
  - A) Encryption
  - ☒ B) Digital signature
  - C) Compression
  - D) Routing
- 40. What is a hash collision?
  - A) Same input
  - ☒ B) Same output for different inputs
  - C) Encryption
  - D) Compression
- 41. Which hash is strongest?
  - A) MD5
  - B) SHA-1
  - ☒ C) SHA-256
  - D) CRC
- 42. What is a one-way function?
  - A) Reversible
  - ☒ B) Irreversible
  - C) Fast
  - D) Small
- 43. What ensures authentication in PKI?
  - A) Encryption
  - ☒ B) Certificates

- C) Compression
- D) Routing
- 22. Which protocol uses tickets for authentication?
  - A) SSL
  - ☒ B) Kerberos
  - C) FTP
  - D) HTTP
- 23. What is a session key?
  - A) Permanent key
  - ☒ B) Temporary key
  - C) Public key
  - D) Private key
- 24. What is mutual authentication?
  - A) One-way authentication
  - ☒ B) Both parties verify each other
  - C) Anonymous login
  - D) Weak authentication
- 25. What ensures freshness in Kerberos?
  - A) Encryption
  - ☒ B) Timestamp
  - C) Compression
  - D) Hashing
- 26. What is a replay attack?
  - A) Data modification
  - ☒ B) Resending valid data
  - C) Encryption
  - D) Compression
- 27. What is the function of an authenticator in Kerberos?
  - A) Data storage
  - ☒ B) Identity proof
  - C) Encryption
  - D) Compression
- 28. What happens if system clocks differ in Kerberos?
  - A) Faster authentication
  - ☒ B) Failure
  - C) Better security
  - D) No effect
- 29. What is the KDC composed of?
  - A) Client & Server
  - ☒ B) AS & TGS
  - C) Firewall & IDS
  - D) Router & Switch
- 30. What is the AS in Kerberos?
  - A) Application Server
  - ☒ B) Authentication Server
  - C) Access Server
  - D) Admin Server
- 31. What is X.509 certificate format used in?
  - A) Networking
  - ☒ B) Digital certificates
  - C) Compression
  - D) Routing
- 32. What identifies the certificate owner?
  - A) Public key
  - ☒ B) Subject field

- C) Compression
  - D) Routing
44. What is key escrow?
- A) Key deletion
  - ☒ B) Key storage by third party
  - C) Encryption
  - D) Compression
45. What is identity spoofing?
- A) Encryption
  - ☒ B) Fake identity
  - C) Compression
  - D) Routing
46. What is trust anchor?
- ☒ A) Root CA
  - B) Client
  - C) Server
  - D) Router
47. What ensures integrity of message?
- A) Encryption
  - ☒ B) Hash
  - C) Compression
  - D) Routing
48. What is digital envelope?
- A) Hash
  - ☒ B) Encrypted message + key
  - C) Compression
  - D) Routing
49. What is certificate fingerprint?
- A) Encryption
  - ☒ B) Hash of certificate
  - C) Compression
  - D) Routing
50. What is PKI used for?
- A) Routing
  - ☒ B) Secure communication
  - C) Compression
  - D) Storage